Enhancing Smartwatch Health Applications through Federated Learning and Cross-Functional Data Analysis

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Abstract

This paper presents a digital health coach that leverages wearable technology while ensuring data sovereignty through federated learning (FL). While most health applications process data independently, the proposed system mixes data from different categories. Incorporating heart rate, sleep, physical activity, and mental health data into menstrual cycle tracking produces more robust predictions, particularly for irregular cycles. To overcome the lack of public datasets, synthetic human cohort data was generated thanks to a probabilistic model. An LSTM network for predicting cycle length, ovulation day, and cycle phase was then implemented and developed under centralized and FL paradigms. The centralized model achieved a 1 day Mean Absolute Error for cycle length and 93% accuracy for phase prediction. While the FL approach demonstrated feasibility, performance decreased due to computation limitations. Ultimately, this research confirms that cross-functional, privacy-preserving health coaching is possible and achieves better results than traditional digital coaches.

Keywords: Artificial Intelligence, Health, Wearables, Federated Learning, Menstrual Cycle Prediction

1 Introduction

In recent years, there has been an increasing interest in employing artificial intelligence (AI) in healthcare, particularly through the use of wearable devices. As the prevalence of chronic diseases rises and access to healthcare becomes constrained, digital health coaches have emerged as vital interactive systems to assist individuals. However, these applications face two major challenges: the isolation of health metrics into separate categories and the critical need for user data privacy. This extended abstract presents the research work conducted to develop a privacy-preserving smartwatch application that cross-analyzes continuous health and lifestyle data collected via wearable sensors.

2 Related Work

Current health applications can be categorized into native smartphone apps, Large Language Model (LLM) assistants, and wearable-specific applications. Even though they offer extensive tracking, they often process data in isolated categories.

For menstrual cycle tracking, most platforms rely solely on averages and manual user inputs, such as average cycle length, average period duration, and the date of the last period. To achieve more advanced predictions, various algorithms can be employed, including decision trees, ARIMA models, and Convolutional Neural Networks (CNNs). However, Long Short-Term Memory (LSTM) networks have proven to be the most accurate [1]. To date, effective LSTM models mainly rely on basal body temperature and heart rate. Yet, no existing model incorporates an individual's mental health, sleep quality, and physical activity levels, even though research demonstrates that these indicators influence menstrual cycles and help identify the current cycle phase.

To address privacy concerns, Federated Learning (FL) has emerged as a state-of-the-art method where training data remains isolated on user devices. Moreover, there are frameworks like Flower that can be used on mobile platforms like Android or iOS, making them suitable for mobile applications [2][3]. Despite these advancements, applying FL to cross-functional wearable data remains underexplored. This project builds upon FL to interconnect these new variables while ensuring strict user data confidentiality.

3 Details of Research

3.1 Health Application

The aim of this study is to propose a Health Guardian leveraging wearables and ensuring data privacy through federated learning. The application offers five functionalities: heart rate monitoring, physical activity tracking, sleep analysis, mental health assessment, and menstrual cycle tracking. While existing applications transmit data to distant servers, this system retains user data locally via federated learning,

using the Flower paradigm [4]. By combining data from different categories, such as sleep quality, mental wellness, physical activity levels, and heart rate, the goal is to make menstrual cycle predictions more robust, especially for irregular cycles. The rest of this section focuses on these specific predictions.

3.2 Synthetic Data Generation

The absence of comprehensive public menstrual cycle datasets necessitated the generation of synthetic data. A Python script utilizing probabilistic models was developed to simulate physiological and behavioral user profiles, generating both regular and irregular cycles with dynamic transitions over time. Selected indicators either signify the current cycle phase, such as basal body temperature and heart rate fluctuations [5], or directly impact cycle regularity. For instance, high stress, depression and excessive physical activity were modeled to induce irregular cycles [6][7], while sleep quality and activity levels decrease prior to and during menstruation [8]. The resulting dataset provides daily continuous values for temperature, heart rate, physical activity, sleep, and mental health scores across multiple consecutive cycles.

3.3 Menstrual Cycle Prediction

The objective of the menstrual cycle tracker is to predict the remaining days until the next menstruation, the ovulation day, and the current cycle phase. The proposed architecture employs a bidirectional Long Short-Term Memory (LSTM) network, optimized through hyperparameter fine-tuning to process time-series health data accurately. The predictive model utilizes daily cross-functional inputs: average cycle length, average period length, menstruator regularity status, basal body temperature, resting heart rate, stress levels, depression (PHQ-9) and anxiety (GAD-7) scores, sleep quality, and physical activity scores. Within the user interface, individuals can consult their cycle history and view these predictive insights directly on a dedicated calendar tab, as illustrated in Figure 1.

4 Results and Conclusion

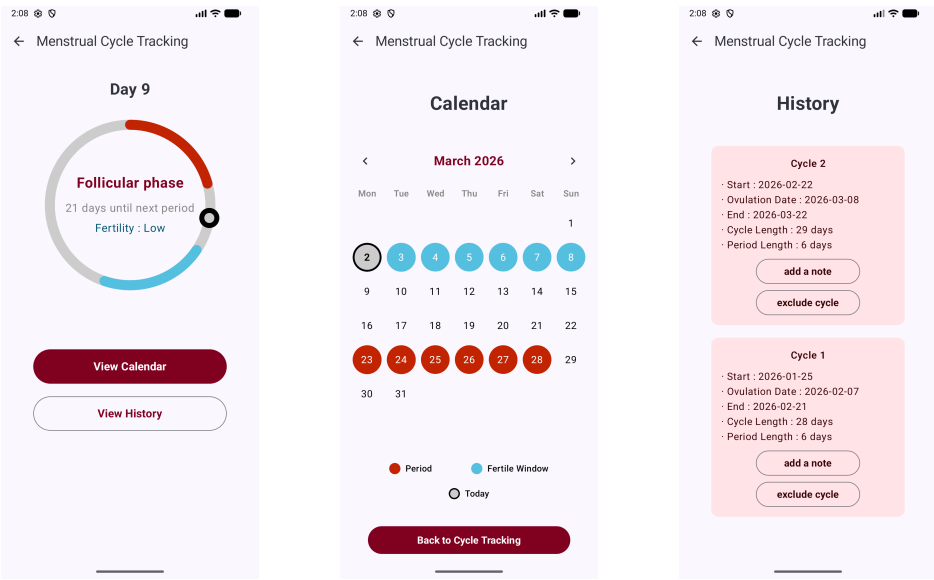
Experiments were conducted on three datasets: a representative dataset containing 25% irregular menstruators, a dataset with only regular menstrual cycles and a dataset with only irregular cycles.

Under centralized learning, the proposed model achieved high accuracy. On the representative dataset, the Mean Absolute Error (MAE) for cycle length was 1.00 day, with a 93.91% phase prediction accuracy. The model also performed well on strictly regular cycles (MAE 0.66 days) and strictly irregular cycles (MAE 1.63 days).

When applying federated learning, performance decreased, resulting in a cycle MAE of 2.26 days on the representative dataset. This decline is attributed to severe computation limitations, as only 16 edge clients could be simulated.

Additionally, a naive version of the model which excludes sleep, physical activity, and mental health scores performed slightly worse across centralized evaluations and struggled significantly more during federated learning on the irregular dataset. This comparison highlights the benefit of integrating lifestyle factors to improve menstrual cycle predictions.

Ultimately, this research demonstrates the successful development of a digital health application that



(a) Tracker (b) Calendar (c) History

Figure 1: User Interface of the Menstrual Cycle Tracking Functionality

effectively combines data from different categories and utilizes federated learning. Furthermore, it confirms that cross-functional data analysis improves prediction accuracy for irregular menstruators, validating this approach for privacy-preserving health coaching.

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